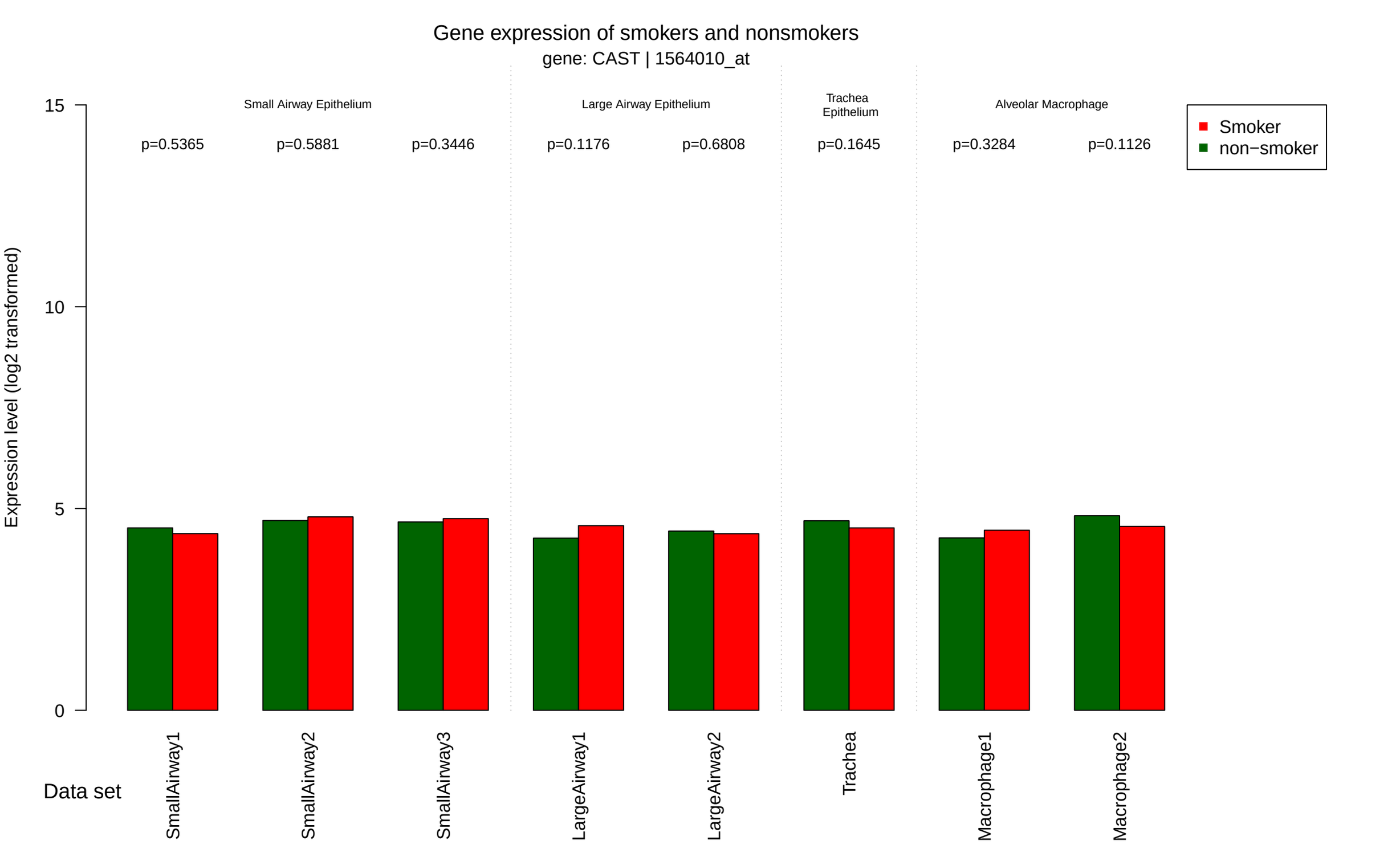
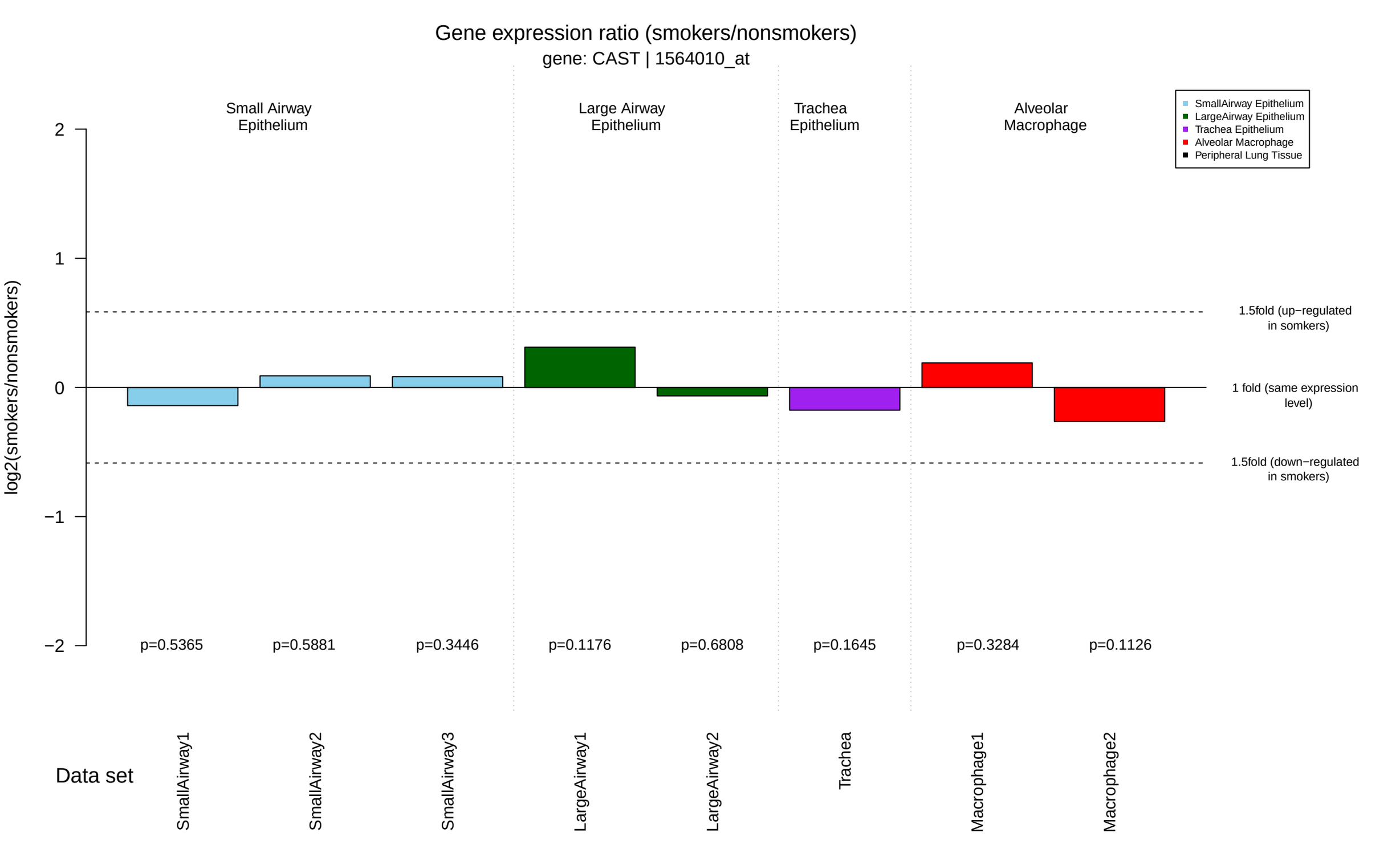
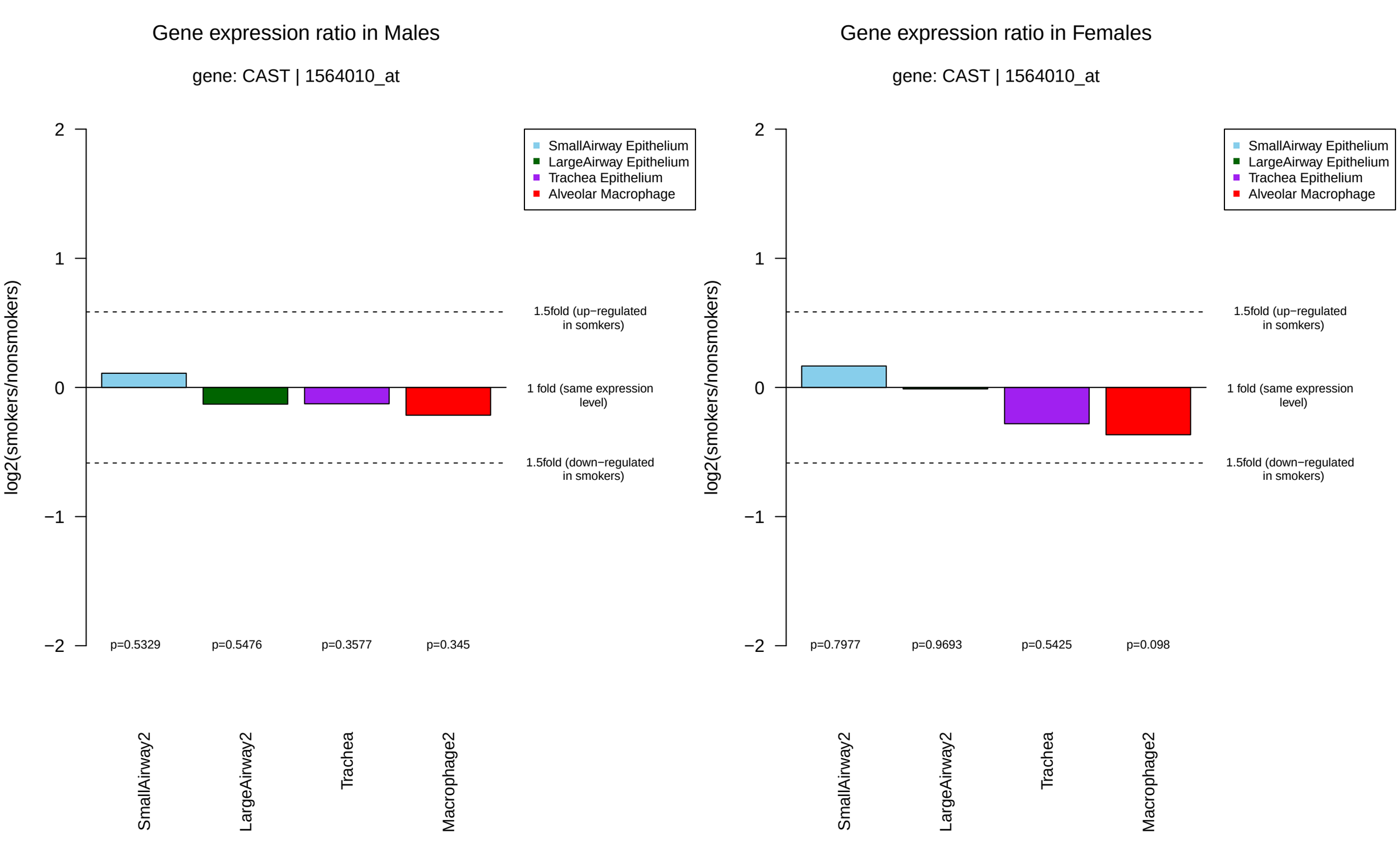
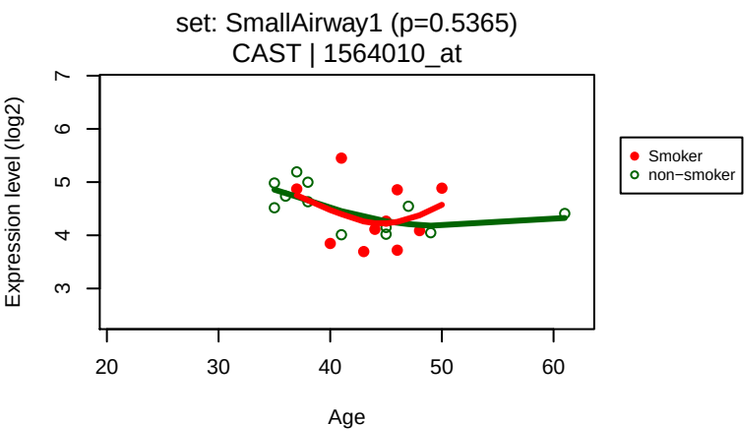
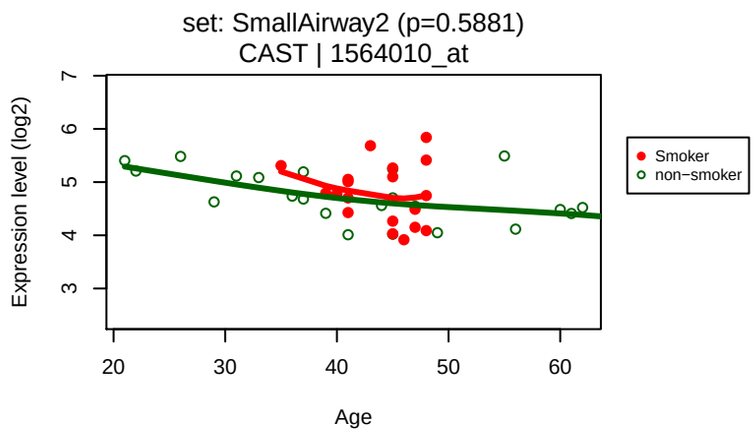
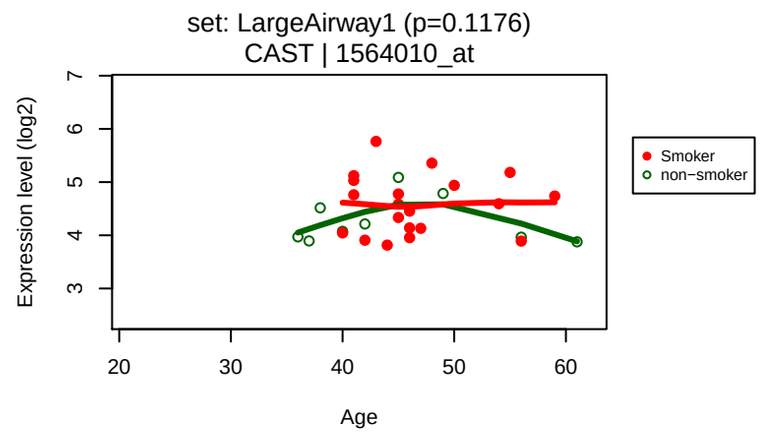

Gene expression of smokers and nonsmokers
gene: CAST | 1564010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5365
p=0.5881
p=0.3446
p=0.1176
p=0.6808
p=0.1645
p=0.3284
p=0.1126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

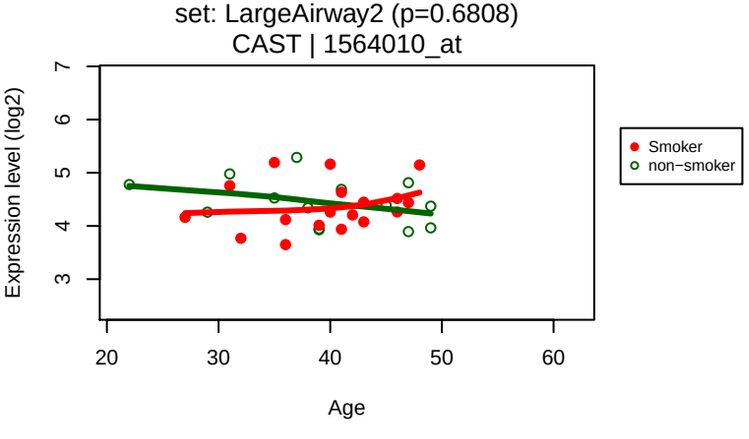

Gene expression ratio (smokers/nonsmokers)
gene: CAST | 1564010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5365
p=0.5881
p=0.3446
p=0.1176
p=0.6808
p=0.1645
p=0.3284
p=0.1126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

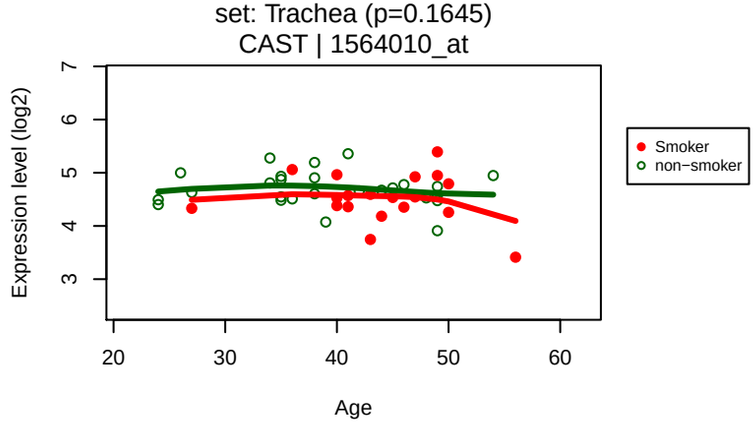

Gene expression ratio in Males
gene: CAST | 1564010_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5329
p=0.5476
p=0.3577
p=0.345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAST | 1564010_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7977
p=0.9693
p=0.5425
p=0.098
SmallAirway2
LargeAirway2
Trachea
Macrophage2

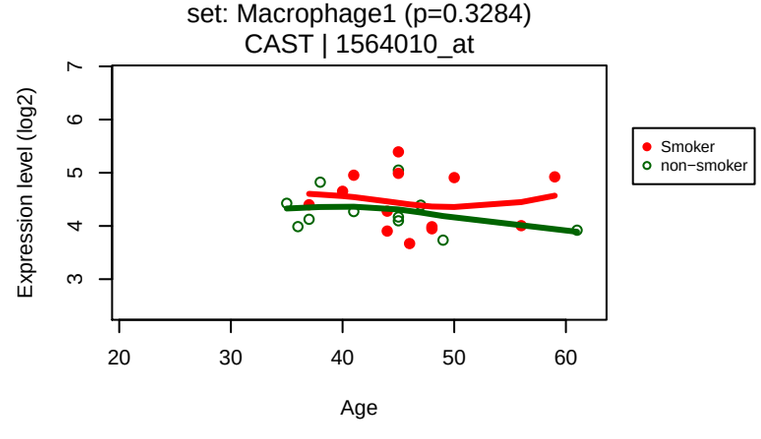

set: SmallAirway1 (p=0.5365)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker

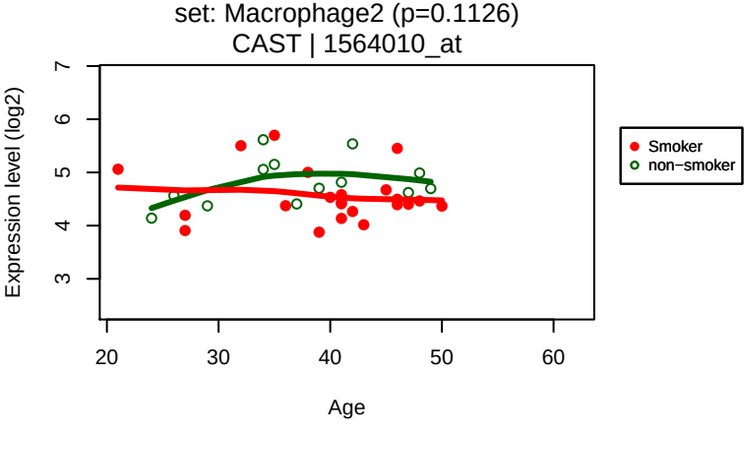

set: SmallAirway2 (p=0.5881)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker

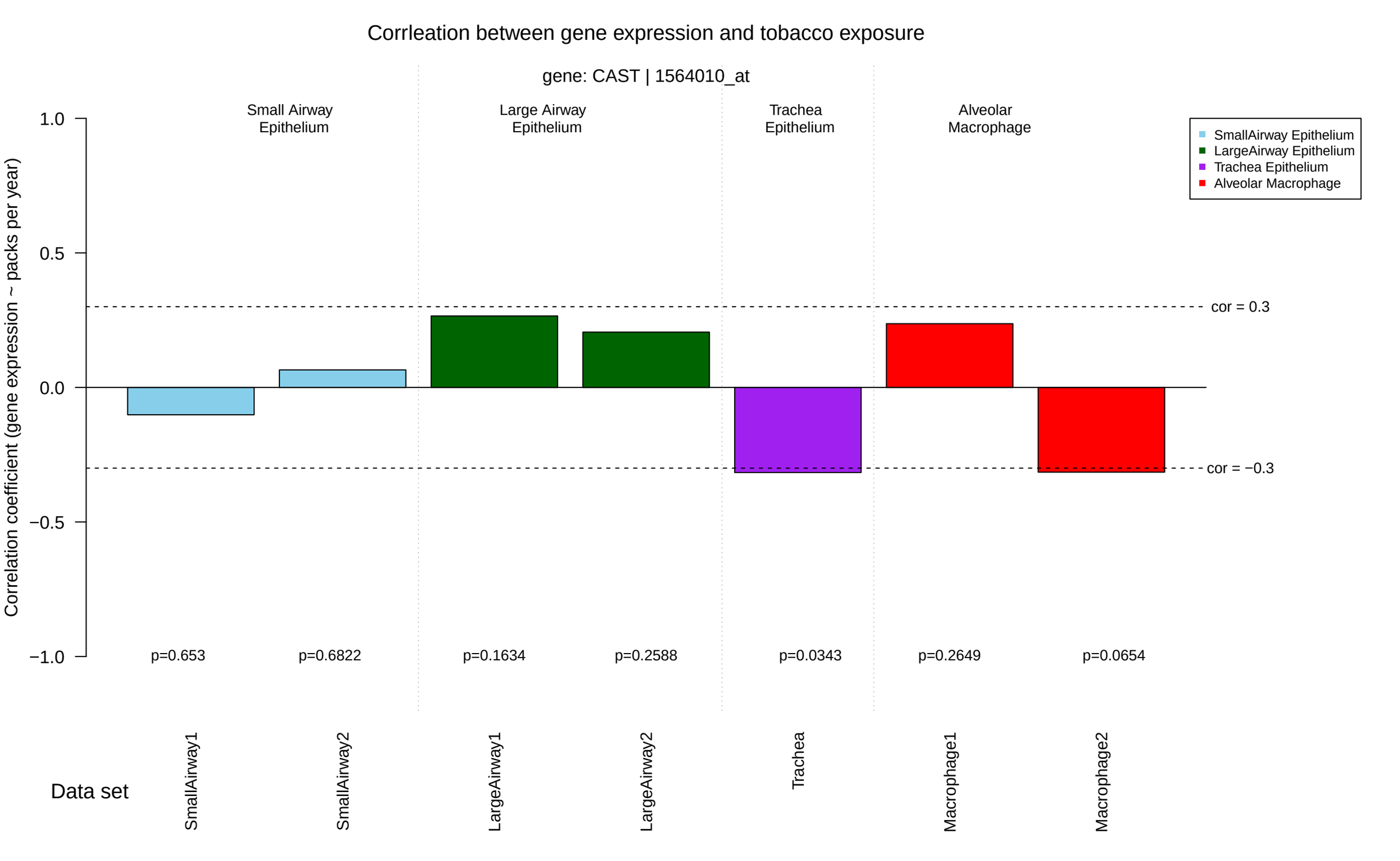

set: LargeAirway1 (p=0.1176)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6808)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1645)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3284)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1126)
CAST | 1564010_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAST | 1564010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.653
p=0.6822
p=0.1634
p=0.2588
p=0.0343
p=0.2649
p=0.0654
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2